

Aryan Sharma
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Dear Hiring Manager,

I am writing to express my interest in the Applications Programmer Associate position at the University of Michigan. With a background in software engineering and data science, I am enthusiastic about contributing to a leading research institution known for its innovative community.

In my recent role as a Software Engineer at AVNI Healthcare, I developed a scalable web-based Training Management System, utilizing Next.js, MongoDB, and AWS for complex needs of global programs. My experience at Amazon, where I engineered solutions using AWS DynamoDB to eliminate system conflicts, has sharpened my skills in debugging and writing reliable code—key requirements for this position.

I am proficient in several programming languages including Python and JavaScript, as well as adept in developing custom web applications—bringing the required technical qualifications mentioned. My academic training complements my practical experience, equipping me with a solid foundation to contribute effectively to your development team.

I am excited about the opportunity to contribute to the University of Michigan's Robotics department and am eager to discuss how my skills can align with your needs. I welcome the opportunity for an interview to further discuss my qualifications.

Thank you for considering my application.

Sincerely,
Aryan Sharma

ARYAN SHARMA

✉ aryanraj@umich.edu ☎ 7344892551 📍 Austin, TX 📁 Portfolio 🌐 LinkedIn 🐙 GitHub

EDUCATION

- University of Michigan**, *Master of Science in Data Science* 📄 08/2023 – 05/2025 | Ann Arbor, US
• Relevant Coursework: Machine Learning, Large Language Models, Information Retrieval, Database Management Systems.
- PES University**, *Bachelor of Technology in Computer Science and Engineering* 08/2019 – 06/2023 | Bengaluru, IN
• Prof. M R Doreswamy Merit Scholarship Recipient for excellent academic performance.
• Relevant Coursework: Machine Learning, Statistics for Data Science, Cloud Computing, Big Data, Data Analytics.

PROFESSIONAL EXPERIENCE

- AVNI Healthcare Inc.(Consultant for Ansrsource)**, *Software Engineer* 📄 07/2025 – Present | Remote, US
• Developing **AWS-hosted** Training Management System (TMS) to automate cohort scheduling and enrollment, targeting **10x efficiency improvement** for L&D administrators.
• Building web-based platform using **Next.js**, **MongoDB**, and **AWS Lambda** supporting enterprise-scale onboarding and leadership development programs across multiple geographies.
• Implementing automated notification systems using **API Gateway** and tracking capabilities, eliminating manual processes for compliance driven organizations.
- Amazon**, *Software Development Engineer Intern* 📄 05/2024 – 08/2024 | Seattle, US
• Engineered distributed locking mechanism using **AmazonDynamoDB**, eliminating 100% of concurrent **AWS Batch Scheduler** conflicts and reducing vCPU calculation errors by 65%.
• Built operational monitoring system with **AWS CloudWatch**, reducing troubleshooting time by 30% and improving system reliability by 5%.
• Collaborated with cross-functional teams to **optimize batch processing workflows** for enterprise-scale applications.
- Genpact**, *Data Scientist* 📄 02/2023 – 08/2023 | Bengaluru, IN
• Developed customer segmentation models using **Python (Scikit-learn, Pandas)** and **A/B testing**, reducing churn by 30% and increasing average order value by 25%.
• Created interactive **Tableau** dashboards for executive decision-making, tracking **KPIs** across customer lifecycle.
• Applied **causal inference** methodologies to identify key churn drivers and optimize retention strategies.
- Smarthub.ai**, *Machine Learning Engineer Intern* 📄 06/2022 – 10/2022 | Bengaluru, IN
• Built **linear regression** model for **robotic** arm optimization in automated vehicle painting for **TVS Motors**.
• Achieved 98% reduction in paint wastage through **predictive modeling**, significantly reducing operational costs.
- Skan.ai**, *Data Science Intern* 📄 06/2021 – 08/2021 | Bengaluru, IN
• Analyzed large-scale operational logs using **SQL** and **MongoDB** to identify process redundancies.
• Delivered insights that reduced operational redundancies by 30% and improved team productivity by 20%.

PUBLICATIONS

- RtTSLC: A Framework for Real-Time Two-Handed Sign Language Translation**, 06/2023
Springer, DOI: 10.1007/978-981-99-0769-4_62 📄
• Developed **CNN-based** framework achieving 98% accuracy with <100ms latency for **real-time sign language translation**.
- Sign Language Translation Systems: A Systematic Literature Review**, 10/2022
IGI Global, DOI: 10.4018/IJSI.311448 📄
• **Synthesized 200+ papers**, identifying 5 research gaps and defining 4 novel methodological approaches.
- Cardiac Anomaly Detection Models for Wearable Devices**, 10/2021
IEEE HiPC Student Research Symposium
• Presented **deep learning architecture** achieving 98% accuracy in cardiac anomaly detection from **2-lead ECG signals**.

PROJECTS

- GDINOSAUR**, *Grounding DINO with Spatial Awareness* 01/2024 – 05/2024
• Generated 7,000+ synthetic image dataset using **Blender** for 30 COCO categories.
• Improved Grounding DINO's spatial understanding by **15% mAP** through **fine-tuning** with Elastic Weight Consolidation.
- ECG Classification**, *CHIPS Research Lab* 08/2021 – 12/2021
• Developed **TensorFlow**-based model for cardiac condition classification from wearable ECG data.
• Achieved 98% classification accuracy, enabling real-time cardiac monitoring capabilities.

SKILLS

Programming Languages: Python, R, Java, C++, SQL, JavaScript, TypeScript, Go, **Machine Learning:** TensorFlow, PyTorch, Keras, Scikit-learn, Hugging Face, NumPy, Pandas, **Cloud & Big Data:** AWS (EC2, Lambda, DynamoDB, CloudWatch), GCP, Azure, Spark, Hadoop, Kafka, Airflow, **Databases:** MySQL, MongoDB, PostgreSQL, Cassandra, DynamoDB, Snowflake, **Data Visualization:** Tableau, Power BI, AWS QuickSight, Matplotlib, Seaborn, **Web Development:** React, Vue.js, HTML/CSS, REST APIs, Docker, Kubernetes